

The obstruction to a skew-brace bicrossed decomposition is a *bilinear* second-cohomology class

The additive forgetful map descends; the product forgetful map is not injective (the W_4 separation)

MacBeth
Kodamai

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Recipient: Rick (peer review); internal (Kodamai; Neil Ghani / Robin Langer).

Scope: kernel ideal D a *trivial brace*; base ring Rathee–Yadav H_{sb}^2 machinery (arXiv:2601.12371).

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Trust grade (honest)

proved for the whole cohomological package below, *contingent on two clearly separated inputs*:

- (a) Task 1 (well-definedness of the additive forgetful map φ_+) is proved here from first principles at the cochain level, **airtight, with no extra hypothesis** beyond the standing trivial-brace-kernel hypothesis (§2).
- (b) The numerical split-triple counts for W_1 – W_4 on $\mathbb{Z}/8$, $\mathbb{Z}_2^2$, $\mathbb{Z}_2 \times \mathbb{Z}_4$ (and in particular $W_4 = (\text{mult}, \text{add}, \text{sb}) = (Y, Y, N)$) are imported as **computed** from the enumerator (scratch file 2026-09-25-kernel-triviality-check.md); they are not re-derived here.

The *cohomological identification* of these computed splits with the vanishing of the three classes $[\Omega], [\tau], [\beta]$ is the new **proved** content. The **non-trivial-brace-kernel** instances on $\mathbb{Z}_2 \times \mathbb{Z}_4$ (**sb#11**, **sb#15**; kernel additively $\mathbb{Z}/4$ but internally a non-trivial brace) are **OPEN** and lie outside Rathee–Yadav; see §5.

1 Setup and standing hypotheses

Let $(G, +, \circ)$ be a skew (left) brace: $(G, +)$ and $(G, \circ)$ are groups and $a \circ (b + c) = (a \circ b) - a + (a \circ c)$. Write $\lambda_a(b) = -a + (a \circ b)$, so $a \circ b = a + \lambda_a(b)$. Fix an *ideal* $D \trianglelefteq G$ and the induced extension

$$0 \longrightarrow D \xrightarrow{i} G \xrightarrow{\pi} H := G/D \longrightarrow 0. \quad (1)$$

We take H as the acting skew brace and $D =: I$ as the kernel, following the notation of Rathee–Yadav (arXiv:2601.12371, hereafter **RY**).

Standing hypothesis (SH). D is a *trivial brace*: $(D, +)$ abelian and $a \circ b = a + b$ on D . Then H acts on D via a *good triplet* (ν, μ, σ) determined by any set-theoretic st-section $s : H \rightarrow G$ ($s(0) = 0$, $\pi s = \text{id}_H$):

$$\nu_h(x) = \lambda_{s(h)}(x), \quad \mu_h(x) = -s(h) + x + s(h), \quad \sigma_h(x) = s(h)^{-1} \circ x \circ s(h),$$

where $\nu : (H, \circ) \rightarrow \text{Aut}(D, +)$ is a homomorphism and $\mu : (H, +) \rightarrow \text{Aut}(D, +)$, $\sigma : (H, \circ) \rightarrow \text{Aut}(D, +)$ are anti-homomorphisms (RY §2). These actions are independent of the choice of s . All four canonical witnesses W_1 – W_4 have trivial-brace kernels, so (SH) holds and RY applies directly (scratch file, this session).

RY second cohomology of a skew brace

To an st-section s RY associate the pair of 2-cochains

$$\beta(h_1, h_2) := -s(h_1 + h_2) + s(h_1) + s(h_2), \quad (2)$$

$$\tau(h_1, h_2) := -s(h_1 \circ h_2) + (s(h_1) \circ s(h_2)) = \nu_{h_1 \circ h_2}(\bar{\tau}(h_1, h_2)), \quad (3)$$

with $\bar{\tau}(h_1, h_2) := s(h_1 \circ h_2)^{-1} \circ s(h_1) \circ s(h_2)$. RY prove (their eqn. following (3.9), citing Lebed–Vendramin / Rathee–Yadav [20]) that $(\beta, \tau) \in Z_{\text{Sb}}^2(H, D)$ iff

$$\begin{aligned} \mu_{\lambda_{h_1}(h_3)}(\tau(h_1, h_2)) - \tau(h_1, h_2 + h_3) + \tau(h_1, h_3) &= \nu_{h_1}(\beta(h_2, h_3)) + \mu_{h_1 \circ h_3}(\beta(h_1, -h_1)) \\ &\quad - \beta(-h_1, h_1 \circ h_3) - \beta(h_1 \circ h_2, \lambda_{h_1}(h_3)), \end{aligned} \quad (3.10)$$

together with $\beta \in Z_{\text{Gp}}^2((H, +), D; \mu)$ and $\bar{\tau} \in Z_{\text{Gp}}^2((H, \circ), D; \sigma)$. Equation (3.10) is the *bilinear coupling*: it is a genuine constraint tying β to τ . The group $H_{\text{Sb}}^2(H, D) = Z_{\text{Sb}}^2/B_{\text{Sb}}^2$ classifies brace extensions (1) inducing the fixed triplet (ν, μ, σ) (RY §3), the zero class being the split (bicrossed) extension.

2 Task 1 (crux): the additive forgetful map descends

Definition 1. $\varphi_+ : H_{\text{Sb}}^2(H, D) \longrightarrow H_{\text{Gp}}^2((H, +), D; \mu), \quad [(\beta, \tau)] \longmapsto [\beta]$.

Well-definedness is the statement that a *brace coboundary*—the shift of (β, τ) under a change of normalised st-section—shifts the β -component by an *additive* group 2-coboundary, so that $[\beta]$ is independent of the representative. The danger, flagged explicitly, is that the (3.10) coupling might inject τ -dependent terms into the β -shift that are *not* additive coboundaries. We prove it does not.

The coboundary group B_{Sb}^2 (parametrisation of section change). Any two normalised st-sections s, s' of (1) have $\pi s = \pi s' = \text{id}_H$, so $s'(h)$ and $s(h)$ lie in the same fibre and differ by an element of D . Setting

$$\theta(h) := -s(h) + s'(h) \in D, \quad \text{equivalently} \quad s'(h) = s(h) + \theta(h), \quad \theta(0) = 0, \quad (4)$$

parametrises *all* changes of normalised st-section by an arbitrary $\theta : H \rightarrow D$ with $\theta(0) = 0$. By definition $B_{\text{Sb}}^2(H, D)$ is the set of shifts $(\beta' - \beta, \tau' - \tau)$ realised as θ ranges over such maps.

Lemma 2 (Additive component of a brace coboundary). *Under (4), with β, β' the additive cochains (2) of s, s' ,*

$$(\beta' - \beta)(h_1, h_2) = \mu_{h_2}(\theta(h_1)) - \theta(h_1 + h_2) + \theta(h_2) = (d\theta)(h_1, h_2), \quad (5)$$

*the standard normalised group 2-coboundary for the additive group $(H, +)$ acting on $(D, +)$ via the right action μ . In particular the shift is **independent of τ , of ν and of σ** .*

Proof. Compute in the (possibly non-abelian) group $(G, +)$, using only associativity and the definition $\mu_h(y) = -s(h) + y + s(h)$; every displayed term below lies in the abelian normal subgroup D .

$$\begin{aligned}
\beta'(h_1, h_2) &= -s'(h_1 + h_2) + s'(h_1) + s'(h_2) \\
&= -(s(h_1 + h_2) + \theta(h_1 + h_2)) + (s(h_1) + \theta(h_1)) + (s(h_2) + \theta(h_2)) \\
&= -\theta(h_1 + h_2) + \underbrace{(-s(h_1 + h_2) + s(h_1))}_{= \beta(h_1, h_2) - s(h_2)} + \theta(h_1) + s(h_2) + \theta(h_2) \\
&= -\theta(h_1 + h_2) + \beta(h_1, h_2) + (-s(h_2) + \theta(h_1) + s(h_2)) + \theta(h_2) \\
&= -\theta(h_1 + h_2) + \beta(h_1, h_2) + \mu_{h_2}(\theta(h_1)) + \theta(h_2).
\end{aligned}$$

The regrouping in the third line is pure associativity: $-s(h_1 + h_2) + s(h_1) = \beta(h_1, h_2) - s(h_2)$ from (2), and inserting it leaves the string $\cdots + \beta(h_1, h_2) - s(h_2) + \theta(h_1) + s(h_2) + \dots$, whose middle block $-s(h_2) + \theta(h_1) + s(h_2) = \mu_{h_2}(\theta(h_1))$. All five surviving summands lie in the abelian group $(D, +)$, so they commute and

$$(\beta' - \beta)(h_1, h_2) = \mu_{h_2}(\theta(h_1)) - \theta(h_1 + h_2) + \theta(h_2). \quad \square$$

Theorem 3 (Task 1). φ_+ is well-defined: if (β_1, τ_1) and (β_2, τ_2) are cohomologous in $Z_{\text{Sb}}^2(H, D)$, then $[\beta_1] = [\beta_2]$ in $H_{\text{Gp}}^2((H, +), D; \mu)$. Consequently φ_+ is a group homomorphism. **No hypothesis beyond (SH) is needed; the compatibility (3.10) does not obstruct.**

Proof. Cohomologous means $(\beta_1, \tau_1) - (\beta_2, \tau_2) \in B_{\text{Sb}}^2$, i.e. realised by some $\theta : H \rightarrow D$ with $\theta(0) = 0$ via a change of st-section (4). By Lemma 2 the additive component is $\beta_1 - \beta_2 = d\theta \in B_{\text{Gp}}^2((H, +), D; \mu)$. Hence $[\beta_1] = [\beta_2]$. Additivity of φ_+ is immediate from $\beta_{(\beta_1 + \beta_2)} = \beta_1 + \beta_2$ (RY note the association $(\beta, \tau) \mapsto \beta$ is $\mathbb{Z}$ -linear on cochains). $\square$

Remark 4 (Why the coupling does not leak, and contrast with $\varphi_\circ$). The point is structural: (2) is *verbatim* the additive group 2-cocycle of the underlying extension $0 \rightarrow (D, +) \rightarrow (G, +) \rightarrow (H, +) \rightarrow 0$, and μ is exactly its conjugation action; so β never "sees" $\circ$, and its coboundary is forced to be the additive one. RY's multiplicative forgetful map $\varphi_\circ : [(\beta, \tau)] \mapsto [\bar{\tau}]$ into $H_{\text{Gp}}^2((H, \circ), D; \sigma)$ descends by the dual computation (their well-definedness proof, ~line 813): with $\theta_1(h) := \nu_h^{-1}(\theta(h))$,

$$(\bar{\tau}_1 - \bar{\tau}_2)(h_1, h_2) = \theta_1(h_2) - \theta_1(h_1 \circ h_2) + \sigma_{h_2}(\theta_1(h_1)),$$

a σ -twisted multiplicative coboundary—involving no β . Thus the coboundary map

$$\theta \longmapsto (d\theta|_+, d\theta|_\circ) \in B_{\text{Gp}}^2((H, +), D; \mu) \times B_{\text{Gp}}^2((H, \circ), D; \sigma)$$

is *diagonal*: (3.10) couples the *cocycles* but not the *coboundaries*. This is precisely why *both* forgetful maps descend—and, as §4 shows, why their product can still fail to be injective. We derived φ_+ directly, not by symmetry: the additive case is in fact the simpler of the two (no ν -renormalisation), and the manifest decoupling is the content, not an assumption.

Exact coboundary formula used. For the record,

$$B_{\text{Gp}}^2((H, +), D; \mu) : (d\theta)(h_1, h_2) = \mu_{h_2}(\theta(h_1)) - \theta(h_1 + h_2) + \theta(h_2), \quad \theta : H \rightarrow D, \theta(0) = 0,$$

the normalised inhomogeneous coboundary for the right μ -module $(D, +)$; and $\beta' - \beta = d\theta$ by Lemma 2.

3 Task 2: the identification (trivial-kernel case)

Definition 5 (Product forgetful map).

$$\Phi = (\varphi_\circ, \varphi_+) : H_{\text{Sb}}^2(H, D) \longrightarrow H_{\text{Gp}}^2((H, \circ), D; \sigma) \times H_{\text{Gp}}^2((H, +), D; \mu), \quad [(\beta, \tau)] \mapsto ([\bar{\tau}], [\beta]),$$

with $\varphi_\circ$ Rathee–Yadav’s (well-defined, §line 813) and φ_+ ours (Theorem 3).

Theorem 6 (Cohomological identification of the decomposition obstruction). *Assume (SH). Let $[\Omega] := [(\beta, \tau)] \in H_{\text{Sb}}^2(H, D)$ be the class of the extension (1) (with the good triplet (ν, μ, σ) determined by G). Then*

$$[\Omega] = 0 \iff (1) \text{ splits (has a brace-hom section)} \iff G \cong (G/D) \blacktriangleright D \iff D \text{ has a sub-skew-brace complement}$$

Proof. The first equivalence is intrinsic to the RY classification: $H_{\text{Sb}}^2(H, D)$ is in bijection with equivalence classes of brace extensions of H by D inducing (ν, μ, σ) (RY §3), and the zero class is exactly the split/bicrossed extension—so $[\Omega] = 0$ iff (1) is equivalent to the split extension iff it admits a brace-hom section. The remaining equivalences—split $\Leftrightarrow$ internal bicrossed product $\Leftrightarrow$ sub-skew-brace complement—are the registry result **corrected-equivalence-1-2-3** (via Ferri, Thm 3.18), cited, not re-derived. $\square$

Remark 7 (No coprimality). The equivalence “ $[\Omega] = 0 \Leftrightarrow$ split” is purely the classification statement and needs **no** coprimality/Hall/Schur–Zassenhaus hypothesis. Coprimality enters RY only in the much stronger Thm 4.3 (a skew-brace Schur–Zassenhaus: $H_{\text{Sb}}^2(H, D) = 0$ *entirely*, forcing *every* extension to split) when $\gcd(|D|, |H|) = 1$. Our witnesses are the opposite regime— $|D|$ and $|H|$ share primes, $H_{\text{Sb}}^2 \neq 0$, and the *specific* class $[\Omega]$ decides the specific decomposition.

4 Task 3: the W_4 separation (crown)

Let W_4 be the skew brace with $(G, +) = \mathbb{Z}_2 \times \mathbb{Z}_4$ and $D = \{0\} \times \mathbb{Z}/4$, an ideal that is a trivial brace (scratch file). The enumerator gives the split triple

$$(\text{mult-split}, \text{add-split}, \text{sb-split}) = (\text{YES}, \text{YES}, \text{NO}) \quad [\text{computed}].$$

Theorem 8 (Φ is not injective). *For W_4 , $[\Omega] \neq 0$ in $H_{\text{Sb}}^2(H, D)$ while $\Phi([\Omega]) = (0, 0)$. Hence $\ker \Phi \neq 0$: **vanishing of $[\Omega]$ is strictly stronger than simultaneous vanishing of both group Schreier classes $[\bar{\tau}]$ and $[\beta]$.***

Proof. Translate each split through the identification.

- *add-split* = YES: the additive extension $0 \rightarrow (D, +) \rightarrow (G, +) \rightarrow (H, +) \rightarrow 0$ splits, so its class $[\beta] = 0$ in $H_{\text{Gp}}^2((H, +), D; \mu)$; i.e. $\varphi_+([\Omega]) = 0$ (Theorem 3).
- *mult-split* = YES: the multiplicative extension $0 \rightarrow (D, \circ) \rightarrow (G, \circ) \rightarrow (H, \circ) \rightarrow 0$ splits, so $[\bar{\tau}] = 0$ in $H_{\text{Gp}}^2((H, \circ), D; \sigma)$; i.e. $\varphi_\circ([\Omega]) = 0$.
- *sb-split* = NO: by Theorem 6, D has no sub-skew-brace complement $\Rightarrow$ (1) does not split $\Rightarrow [\Omega] \neq 0$.

Therefore $\Phi([\Omega]) = (0, 0)$ with $[\Omega] \neq 0$. $\square$

Corollary 9 (The obstruction is irreducibly bilinear). $[\Omega]$ is not detected by either group cohomology alone, nor by their product: the decomposition obstruction lives in the coupling (3.10) between β and τ , not in $[\beta]$ or $[\bar{\tau}]$. Concretely, $\beta = d\theta_1$ for some additive 1-cochain θ_1 and $\bar{\tau} = d\theta_2$ for some multiplicative 1-cochain θ_2 , yet no single θ trivialises the pair (β, τ) simultaneously; that simultaneous-triviality defect is $[\Omega]$.

Remark 10 (Socle cross-reference: why the separation is possible). RY Thm 3.5 states: if $D = I \leq \text{Soc}(G)$ (equivalently μ trivial and $\nu_h^{-1} = \sigma_h$ for all h) then the brace extension splits iff its associated Λ -group extension does. Thus $I \leq \text{Soc}$ collapses brace-splitting onto group-splitting and forbids a W_4 -type separation. The separation in Theorem 8 therefore requires $D \not\leq \text{Soc}(G)$. This is consistent with, and independently corroborated by, RY’s own Example 3.4/4.19: on the same additive group $\mathbb{Z}_2 \times \mathbb{Z}_4$ the socle has order 2, and RY exhibit an ideal $I \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ (order 4 > $|\text{Soc}|$, hence $I \not\leq \text{Soc}$) whose brace extension does not split even though the Λ -group extension does. Our W_4 ideal $D \cong \mathbb{Z}/4$ is a different ideal of a brace on the same additive group, but the mechanism is identical: $D \not\leq \text{Soc}(G)$ is exactly the failure of the Thm 3.5 hypothesis that permits “both group extensions split, brace does not.” (RY’s Λ -group extension splitting and our componentwise (mult, add)-splittings are logically distinct statements; we invoke Thm 3.5 only for the qualitative “ $I \leq \text{Soc} \Rightarrow$ no separation,” not for a numerical claim.)

5 Open: non-trivial-brace kernels

Hypothesis (SH) is essential: RY’s H_{SB}^2 is built for a *trivial-brace* kernel. On $\mathbb{Z}_2 \times \mathbb{Z}_4$ the enumerator finds 4 skew braces (e.g. **sb#11**, **sb#15**) realising the same (Y, Y, N) obstruction with a kernel that is additively $\mathbb{Z}/4$ but *internally a non-trivial brace* ($\lambda|_D \neq \text{id}$). These lie outside RY and the present identification. Whether an analogous product-forgetful map and bilinear obstruction exist for non-trivial-brace kernels is **OPEN**.

Summary

- T1** $\varphi_+ : [(\beta, \tau)] \mapsto [\beta]$ descends, **proved**, airtight, no extra hypothesis; the β -shift under any change of st-section is the additive coboundary $d\theta$ (Lemma 2), τ -independent. The coupling (3.10) is on cocycles, not coboundaries.
- T2** For trivial-brace D , $[\Omega] = [(\beta, \tau)]$ and $[\Omega] = 0 \Leftrightarrow \text{split} \Leftrightarrow G \cong (G/D) \blacktriangleright D \Leftrightarrow \text{sub-skew-brace complement}$ (Thm 6); no coprimality.
- T3** W_4 : $\Phi([\Omega]) = (0, 0)$ yet $[\Omega] \neq 0$; Φ non-injective; the obstruction is bilinear and requires $D \not\leq \text{Soc}(G)$ (Thm 8, Rem. 10).

Open Non-trivial-brace kernels (**sb#11/15**) outside RY.